Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 23)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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6	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 1 . 1 Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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0 1 . 2 Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 1 . 3 Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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0 2 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

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0 2 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

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0 2 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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0 3 . 1 Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability .....

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Dissolved oxygen .....

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pH .....

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0 3 . 2 Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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0 3 . 3 Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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0 4 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 4 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 5 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 5 . 1 Explain what the standard deviation values tell you about the data.

[2 marks]

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0 5 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

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0 5 . 3 The test gave a probability p = 0.30. State what this means and what the student should conclude.

[2 marks]

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0 5 . 4 Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

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0 5 . 5 Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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0 6 . 1 Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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0 6 . 2 Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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0 6 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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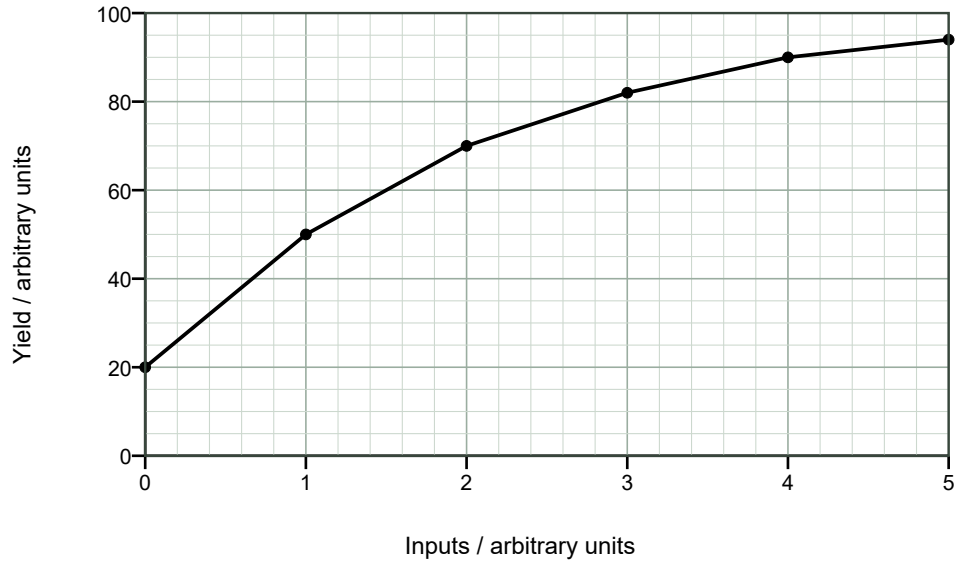
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0 7 Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



0 7 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 7 . 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer ☐ units

0 7 . 3 Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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0 7 . 4 Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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0 7 . 5 Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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0 8 . 1 Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 8 . 2 Describe **five** features of a suitable release site.

[5 marks]

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0 8 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 9 . 1 Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

0 9 . 2 Explain how soil organisms provide services that humans depend on.

[3 marks]

0 9 . 3 Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

0 9 . 4 Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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1 0 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS